

## Linux Practice Tasks - Step by Step Commands (Task 1 to Task 10)

=====

### Task 1: Clear Existing Deployment

=====

#### 1. Stop services:

```
sudo systemctl stop apache2
```

```
sudo systemctl stop mariadb
```

#### 2. Remove Apache:

```
sudo apt remove --purge apache2 apache2-utils apache2-bin apache2.2-common -y
```

#### 3. Remove MariaDB:

```
sudo apt remove --purge mariadb-server mariadb-client mariadb-common -y
```

#### 4. Remove unused packages:

```
sudo apt autoremove -y
```

```
sudo apt autoclean
```

#### 5. Remove application directory:

```
sudo rm -rf /var/www/html/*
```

#### 6. Verify port 80 is free:

```
sudo netstat -tulnp | grep :80
```

=====

### Task 2: Deploy Application Again

=====

#### 1. Update packages:

```
sudo apt update
```

#### 2. Install Apache & MariaDB:

```
sudo apt install apache2 mariadb-server -y
```

#### 3. Start services:

```
sudo systemctl start apache2
```

```
sudo systemctl start mariadb
```

4. Enable services:

```
sudo systemctl enable apache2  
sudo systemctl enable mariadb
```

5. Verify services:

```
sudo systemctl status apache2  
sudo systemctl status mariadb
```

```
=====  
Task 3: Service Health Check Script  
=====
```

1. Create script:

```
nano /home/archana/health_check.sh
```

2. Add below content:

```
#!/bin/bash  
echo "===== Service Health Check ====="  
echo "Date: $(date)"  
echo "Apache Status:"  
systemctl is-active apache2  
echo "MariaDB Status:"  
systemctl is-active mariadb
```

3. Make executable:

```
chmod +x /home/archana/health_check.sh
```

```
archana@ARCHANA:~$ cat service_check.sh  
#!/bin/bash  
echo "===== SERVICE HEALTH CHECK ====="  
echo "Date: $(date)"  
  
echo -n "Apache: "  
systemctl is-active apache2  
  
echo -n "MariaDB: "  
systemctl is-active mariadb  
archana@ARCHANA:~$ chmod 700 service_check.sh  
archana@ARCHANA:~$ ./service_check.sh  
===== SERVICE HEALTH CHECK =====  
Date: Thu Feb 12 12:37:39 IST 2026  
Apache: active  
MariaDB: active  
archana@ARCHANA:~$ |
```

4. Run script:

```
/home/archana/health_check.sh
```

## Task 4: Automate Health Check

1. Create directory:

```
mkdir -p /home/archana/health_logs
```

2. Open crontab:

```
crontab -e
```

3. Add:

```
*/5 * * * * /home/archana/health_check.sh >>  
/home/archana/health_logs/service_status.log
```

4. Verify:

```
crontab -l
```

```
archana@ARCHANA:~$ cat service_check.sh  
#!/bin/bash  
  
LOGDIR="/home/archana/health_logs"  
mkdir -p $LOGDIR  
  
echo "===== SERVICE HEALTH CHECK =====" >> $LOGDIR/service_status.log  
echo "Date: $(date)" >> $LOGDIR/service_status.log  
  
if pgrep apache2 > /dev/null  
then  
    echo "Apache: Running" >> $LOGDIR/service_status.log  
else  
    echo "Apache: Not Running" >> $LOGDIR/service_status.log  
fi  
  
if pgrep mysqld > /dev/null  
then  
    echo "MariaDB: Running" >> $LOGDIR/service_status.log  
else  
    echo "MariaDB: Not Running" >> $LOGDIR/service_status.log  
fi  
  
echo "-----" >> $LOGDIR/service_status.log  
archana@ARCHANA:~$ crontab -e  
No modification made  
archana@ARCHANA:~$ cat ~/health_logs/service_status.log  
===== SERVICE HEALTH CHECK =====  
Date: Thu Feb 12 14:20:01 IST 2026  
Apache: active  
MariaDB: active
```

## Task 5: Log Cleanup Script

1. Create backup directory:

```
mkdir -p /home/archana/log_backup
```

2. Create script:

```
nano /home/archana/log_cleanup.sh
```

3. Add:

```
#!/bin/bash  
SOURCE="/var/log"
```

```

BACKUP="/home/archana/log_backup"
echo "Cleanup started at $(date)"
find $SOURCE -type f -mmin -60 -exec cp {} $BACKUP \;
find $SOURCE -type f -mmin -60 -exec rm {} \;
echo "Cleanup completed at $(date)"

```

4. Make executable:

```
chmod +x /home/archana/log_cleanup.sh
```

```

archana@ARCHANA:~$ cat log_cleanup.sh
#!/bin/bash

LOGDIR="/home/archana/health_logs"
BACKUPDIR="/home/archana/log_backup"

mkdir -p $BACKUPDIR

echo "==== LOG CLEANUP PROCESS ====="
echo "Date: $(date)"

# Find .log files modified in last 60 minutes
FILES=$(find $LOGDIR -type f -name "*.log" -mmin -60)

if [ -z "$FILES" ]; then
    echo "No recent log files found."
else
    for file in $FILES
    do
        echo "Backing up: $file"
        cp "$file" "$BACKUPDIR"

        if [ $? -eq 0 ]; then
            echo "Deleting original: $file"
            rm "$file"
        else
            echo "Backup failed for: $file"
        fi
    done
fi

echo "-----"
archana@ARCHANA:~$ chmod 700 /home/archana/log_cleanup.sh
archana@ARCHANA:~$ /home/archana/service_check.sh
archana@ARCHANA:~$ /home/archana/log_cleanup.sh
==== LOG CLEANUP PROCESS =====
Date: Thu Feb 12 14:32:43 IST 2026
Backing up: /home/archana/health_logs/service_status.log
Deleting original: /home/archana/health_logs/service_status.log
-----

```

## Task 6: Automate Log Cleanup

1. Open crontab:

```
crontab -e
```

2. Add:

```
0 * * * * /home/archana/log_cleanup.sh >> /home/archana/log_backup/cleanup.log
```

```

archana@ARCHANA:~$ cat /home/archana/log_backup/cleanup.log
===== LOG CLEANUP PROCESS =====
Date: Thu Feb 12 19:09:01 IST 2026
Backing up: /home/archana/health_logs/service_status.log
Deleting original: /home/archana/health_logs/service_status.log
=====
===== LOG CLEANUP PROCESS =====
Date: Thu Feb 12 19:10:01 IST 2026
No recent log files found.
=====
===== LOG CLEANUP PROCESS =====
Date: Thu Feb 12 19:11:01 IST 2026
No recent log files found.
=====
===== LOG CLEANUP PROCESS =====
Date: Thu Feb 12 19:12:01 IST 2026
No recent log files found.
=====
===== LOG CLEANUP PROCESS =====
Date: Thu Feb 12 19:13:01 IST 2026
No recent log files found.
=====
===== LOG CLEANUP PROCESS =====
Date: Thu Feb 12 19:14:01 IST 2026
No recent log files found.
=====
===== LOG CLEANUP PROCESS =====
Date: Thu Feb 12 19:15:01 IST 2026
No recent log files found.
=====
===== LOG CLEANUP PROCESS =====
Date: Thu Feb 12 19:16:01 IST 2026
No recent log files found.
=====
===== LOG CLEANUP PROCESS =====
Date: Thu Feb 12 19:17:01 IST 2026
No recent log files found.
=====

```

## =====

### Task 7: Database Backup Script

## =====

1. Create backup directory:

```
mkdir -p /home/archana/db_backup
```

2. Create script:

```
nano /home/archana/db_backup.sh
```

3. Add:

```
#!/bin/bash
```

```
DATE=$(date +"%Y-%m-%d_%H-%M-%S")
```

```
BACKUP_DIR="/home/archana/db_backup"
```

```
mysqldump -u backupuser -p ecomdb > $BACKUP_DIR/ecomdb_$DATE.sql
```

4. Make executable:

```
chmod +x /home/archana/db_backup.sh
```

5. Run:

```
/home/archana/db_backup.sh
```

```

archana@ARCHANA:~$ cat /home/archana/db_backup.sh
#!/bin/bash

DB_NAME="ecomdb"
BACKUP_DIR="/home/archana/db_backups"

mkdir -p $BACKUP_DIR

TIMESTAMP=$(date +%Y-%m-%d_%H-%M-%S)
BACKUP_FILE="$BACKUP_DIR/${DB_NAME}_backup_${TIMESTAMP}.sql"

echo "Starting database backup..."
echo "Date: $(date)"

sudo mysqldump $DB_NAME > $BACKUP_FILE

if [ $? -eq 0 ]; then
    echo "Backup successful: $BACKUP_FILE"
else
    echo "Backup failed!"
fi

archana@ARCHANA:~$ chmod 700 /home/archana/db_backup.sh
archana@ARCHANA:~$ /home/archana/db_backup.sh
Starting database backup...
Date: Thu Feb 12 19:25:07 IST 2026
Backup successful: /home/archana/db_backups/ecomdb_backup_2026-02-12_19-25-07.sql

```

## Task 8: Disk Space Protection Script

1. Create script:

nano /home/archana/disk\_check.sh

2. Add:

```

#!/bin/bash
THRESHOLD=80
USAGE=$(df / | tail -1 | awk '{print $5}' | sed 's/%//g')
if [ $USAGE -gt $THRESHOLD ]; then
    echo "$(date): WARNING - Disk usage above limit" >> /home/archana/disk_warning.log
fi

```

3. Make executable:

chmod +x /home/archana/disk\_check.sh

```

archana@ARCHANA:~$ cat /home/archana/disk_check.sh
#!/bin/bash

LIMIT=80
LOGFILE="/home/archana/disk_warning.log"

USAGE=$(df / | awk 'NR==2 {print $5}' | sed 's/%//')

echo "Checking disk usage..."
echo "Current usage: $USAGE%"

if [ "$USAGE" -gt "$LIMIT" ]; then
    echo "$(date): WARNING - Disk usage is ${USAGE}% (Limit: ${LIMIT}%)" >> $LOGFILE
else
    echo "$(date): Disk usage is normal (${USAGE}%)" >> $LOGFILE
fi

archana@ARCHANA:~$ chmod 700 /home/archana/disk_check.sh
archana@ARCHANA:~$ /home/archana/disk_check.sh
Checking disk usage...
Current usage: 4%
archana@ARCHANA:~$ cat /home/archana/disk_warning.log
Thu Feb 12 19:31:11 IST 2026: Disk usage is normal (4%)
Thu Feb 12 19:31:53 IST 2026: Disk usage is normal (4%)

```

## Task 9: Application Availability Check

### 1. Create script:

nano /home/archana/app\_check.sh

### 2. Add:

```
#!/bin/bash
STATUS=$(curl -s -o /dev/null -w "%{http_code}" http://localhost)
if [ $STATUS -eq 200 ]; then
echo "$(date): Application is UP (HTTP 200)" >> /home/archana/app_status.log
else
echo "$(date): Application is DOWN" >> /home/archana/app_status.log
fi
```

### 3. Make executable:

chmod +x /home/archana/app\_check.sh

### 4. Run:

/home/archana/app\_check.sh

```
archana@ARCHANA:~$ vi /home/archana/app_check.sh
archana@ARCHANA:~$ chmod 700 /home/archana/app_check.sh
archana@ARCHANA:~$ /home/archana/app_check.sh
archana@ARCHANA:~$ cat /home/archana/app_status.log
Thu Feb 12 19:44:46 IST 2026: Application is UP (HTTP 200)
archana@ARCHANA:~$ sudo service apache2 stop
[sudo] password for archana:
archana@ARCHANA:~$ sudo service apache2 status
○ apache2.service - The Apache HTTP Server
   Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor preset: enabled)
   Active: inactive (dead) since Thu 2026-02-12 19:45:46 IST; 9s ago
     Docs: https://httpd.apache.org/docs/2.4/
   Process: 37143 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
   Process: 39172 ExecStop=/usr/sbin/apachectl graceful-stop (code=exited, status=0/SUCCESS)
   Main PID: 37147 (code=exited, status=0/SUCCESS)
    CPU: 1.699s

Feb 12 12:28:31 ARCHANA systemd[1]: Starting The Apache HTTP Server...
Feb 12 12:28:31 ARCHANA systemd[1]: Started The Apache HTTP Server...
Feb 12 19:45:46 ARCHANA systemd[1]: Stopping The Apache HTTP Server...
Feb 12 19:45:46 ARCHANA systemd[1]: apache2.service: Deactivated successfully.
Feb 12 19:45:46 ARCHANA systemd[1]: Stopped The Apache HTTP Server.
Feb 12 19:45:46 ARCHANA systemd[1]: apache2.service: Consumed 1.699s CPU time.
archana@ARCHANA:~$ /home/archana/app_check.sh
archana@ARCHANA:~$ cat /home/archana/app_status.log
Thu Feb 12 19:44:46 IST 2026: Application is UP (HTTP 200)
Thu Feb 12 19:46:04 IST 2026: Application is DOWN (HTTP 000)
archana@ARCHANA:~$ cat cat app_check.sh
cat: cat: No such file or directory
#!/bin/bash

URL="http://localhost"
LOGFILE="/home/archana/app_status.log"

STATUS=$(curl -s -o /dev/null -w "%{http_code}" $URL)

if [ "$STATUS" -eq 200 ]; then
    echo "$(date): Application is UP (HTTP $STATUS)" >> $LOGFILE
else
    echo "$(date): Application is DOWN (HTTP $STATUS)" >> $LOGFILE
fi
```

## Task 10: System Health Report

### 1. Create script:

nano /home/archana/system\_report.sh

### 2. Add:

```
#!/bin/bash
```

```
REPORT="/home/archana/system_health_report.txt"
```

```
echo "System Health Report - $(date)" > $REPORT
```

```
echo "Apache Status: $(systemctl is-active apache2)" >> $REPORT
```

```
echo "MariaDB Status: $(systemctl is-active mariadb)" >> $REPORT
```

```
echo "Port 80 Status:" >> $REPORT
```

```
sudo netstat -tulnp | grep :80 >> $REPORT
```

```
echo "Disk Usage:" >> $REPORT
```

```
df -h >> $REPORT
```

```
echo "Memory Usage:" >> $REPORT
```

```
free -h >> $REPORT
```

### 3. Make executable:

chmod +x /home/archana/system\_report.sh

### 4. Run:

/home/archana/system\_report.sh

```
archana@archana:~$ vi /home/archana/system_report.sh
archana@archana:~$ chmod 700 /home/archana/system_report.sh
archana@archana:~$ cat system_report.sh
#!/bin/bash

REPORT_DIR="/home/archana/system_reports"
mkdir -p $REPORT_DIR

TIMESTAMP=$(date +%Y-%m-%d_%H-%M-%S)
REPORT_FILE="$REPORT_DIR/system_report_$TIMESTAMP.txt"

echo "===== SYSTEM HEALTH REPORT =====" > $REPORT_FILE
echo "Generated on: $(date)" >> $REPORT_FILE
echo "-----" >> $REPORT_FILE

# Apache Status
if pgrep apache2 > /dev/null
then
    echo "Apache Status: Running" >> $REPORT_FILE
else
    echo "Apache Status: Not Running" >> $REPORT_FILE
fi

# MariaDB Status
if pgrep mysqld > /dev/null
then
    echo "MariaDB Status: Running" >> $REPORT_FILE
else
    echo "MariaDB Status: Not Running" >> $REPORT_FILE
fi

# Port 80 Status
if ss -tln | grep -q ":80"
then
    echo "Port 80: Listening" >> $REPORT_FILE
else
    echo "Port 80: Not Listening" >> $REPORT_FILE
fi

echo "-----" >> $REPORT_FILE

# Disk Usage
echo "Disk Usage:" >> $REPORT_FILE
df -h / >> $REPORT_FILE

echo "-----" >> $REPORT_FILE

# Memory Usage
echo "Memory Usage:" >> $REPORT_FILE
free -h >> $REPORT_FILE

echo "-----" >> $REPORT_FILE

echo "Report saved to: $REPORT_FILE"
archana@archana:~$ /home/archana/system_report.sh
Report saved to: /home/archana/system_reports/system_report_2020-02-12_20-02-29.txt
archana@archana:~$ ls /home/archana/system_reports
system_report_2020-02-12_20-02-29.txt
archana@archana:~$ cat /home/archana/system_reports/system_report_*.txt
===== SYSTEM HEALTH REPORT =====
Generated on: Thu Feb 12 20:02:29 IST 2020

Apache Status: Running
MariaDB Status: Not Running
Port 80: Listening
-----
Disk Usage:
Filesystem      Size  Used Avail Use% Mounted on
/dev/sdd        100G   29G  927G   4% /

Memory Usage:
              total        used        free     shared  buff/cache   available
Mem:          7.6Gi       797Mi       6.6Gi       19Mi       811Mi       6.46Gi
Swap:          2.0Gi           0B       2.0Gi
```